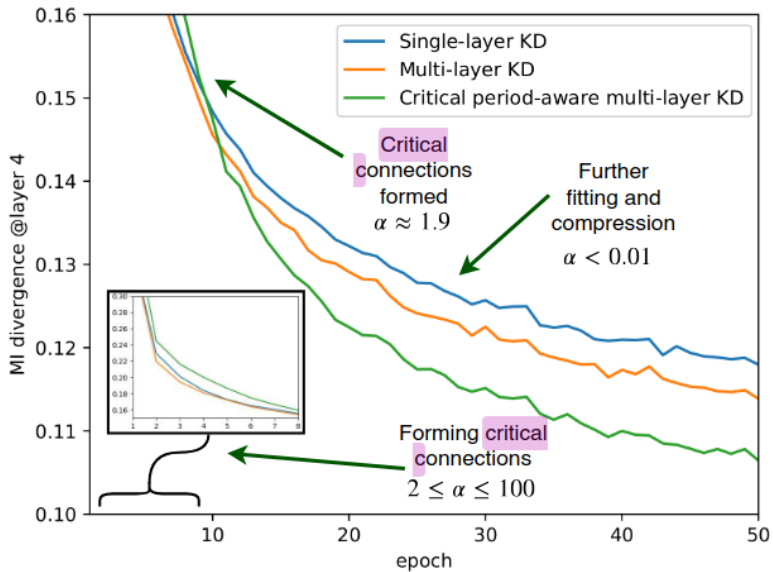


Bayesian multimodeling: multitask learning

MIPT

2025



What is multitask learning?

Multitask and dataset shift?

Linear regression case

$$\mathbf{Y} = \mathbf{X}^T \mathbf{w} + \varepsilon,$$
$$\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{B}^{-1}), \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{A}^{-1}).$$

Task Clustering and Gating for Bayesian Multitask Learning

Consider a 1-layer neural network:

$$\mathbf{y}^i = \mathbf{W}_{\text{task}}^i \sigma(\mathbf{W}_{\text{shared}} \mathbf{x})$$

Can we set an interconnection of tasks?

Task Clustering and Gating for Bayesian Multitask Learning

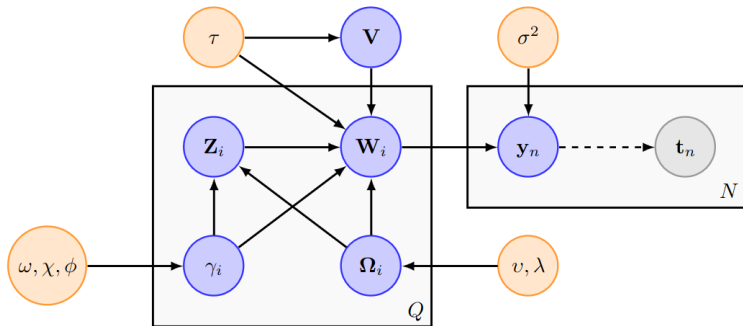
Interconnection of tasks:

- No: $\mathbf{W}_{\text{task}}^i \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$;
- Gaussian mixture: $\mathbf{W}_{\text{task}}^i \sim \sum_j \alpha_j \mathcal{N}(\boldsymbol{\mu}_j, \boldsymbol{\Sigma})$;
- Using gating function: $\mathbf{W}_{\text{task}}^i \sim \sum_j \text{softmax}(\boldsymbol{\alpha})_j \mathcal{N}(\boldsymbol{\mu}_j, \boldsymbol{\Sigma})$;

Sparse Bayesian Multi-Task Learning, 2011

$$\mathbf{y}_n = \mathbf{W}\mathbf{x}_n + \boldsymbol{\mu} + \boldsymbol{\varepsilon}_n;$$

$$\boldsymbol{\varepsilon}_n \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}).$$



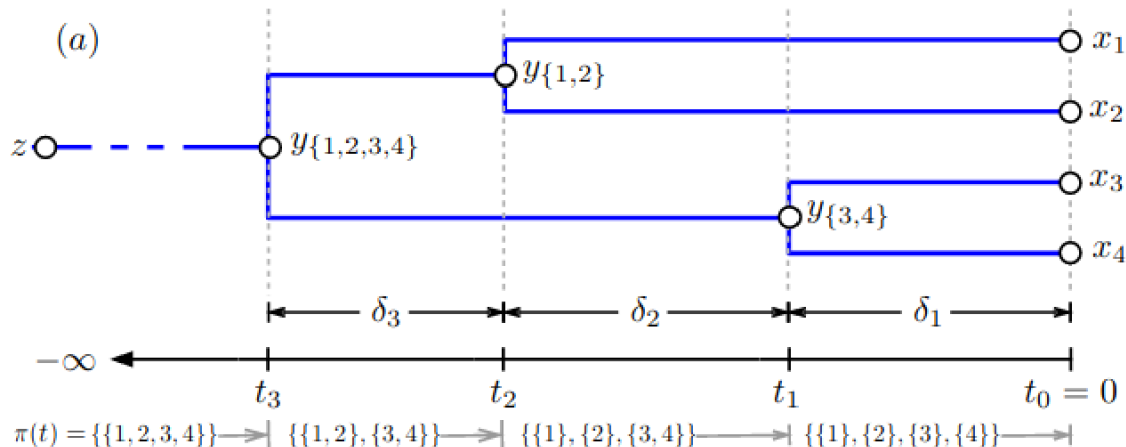
Multitask and domain adaptation?

Bayesian Multitask Learning with Latent Hierarchies

Daume III, 2009

We exploit the intuition that for domain adaptation, we wish to share classifier structure, but for multitask learning, we wish to share covariance structure.

Bayesian Multitask Learning with Latent Hierarchies



Bayesian Multitask Learning with Latent Hierarchies

1. Choose a global *mean* and *covariance* $(\boldsymbol{\mu}^{(0)}, \boldsymbol{\Lambda}) \sim \mathcal{NorTW}(0, \sigma^2 \mathbf{I}, D + 1)$.²
2. Choose a tree structure $(\pi, \boldsymbol{\delta}) \sim \textit{Coalescent}$ over K leaves.
3. For each non-root node i in π (top-down):
 - (a) Choose $\boldsymbol{\mu}^{(i)} \sim \mathcal{Nor}(\boldsymbol{\mu}^{(p_\pi(i))}, \delta_i \boldsymbol{\Lambda})$, where $p_\pi(i)$ is the parent of i in π .
4. For each domain $k \in [K]$:
 - (a) Denote by $\mathbf{w}^{(k)} = \boldsymbol{\mu}^{(i)}$ where i is the leaf in π corresponding to domain k .
 - (b) For each example $n \in [N_k]$:
 - i. Choose input $\mathbf{x}_n^{(k)} \sim \mathcal{D}^{(k)}$.
 - ii. Choose output $y_n^{(k)}$ by:
Regression: $\mathcal{Nor}(\mathbf{w}^{(k)\top} \mathbf{x}_n^{(k)}, \rho^2)$
Classification: $\textit{Bin}(1/(1 + e^{-\mathbf{w}^{(k)\top} \mathbf{x}_n^{(k)}}))$

Bayesian Multitask Learning with Latent Hierarchies

- 1. Choose $\mathbf{R}$ by Eq (2) and deviation covariance $\mathbf{\Lambda} \sim \mathcal{IW}(\sigma^2 \mathbf{I}, D + 1)$.
- 2. Choose a tree structure $(\pi, \delta) \sim \textit{Coalescent}$ over K leaves.
- 3. For each non-root node i in π (top-down):
 - (a) Choose $\mathbf{S}^{(i)} \sim \mathcal{Nor}(\mathbf{S}^{(p_\pi(i))}, \delta_i \mathbf{\Lambda})$, where $p_\pi(i)$ is the parent of i in π .
- 4. For each task $k \in [K]$:
 - (a) Choose $\mathbf{w}^{(k)}$ by (i is the leaf associated with task k): $\mathcal{Nor}(0, (\exp \mathbf{S}^{(i)}) \mathbf{R} (\exp \mathbf{S}^{(i)}))$
 - (b) For each example $n \in [N_k]$:
 - i. Choose input $\mathbf{x}_n^{(k)} \sim \mathcal{D}$.
 - ii. Choose output $y_n^{(k)}$ by:
Regression: $\mathcal{Nor}(\mathbf{w}^{(k)\top} \mathbf{x}_n^{(k)}, \rho^2)$
Classification: $\textit{Bin}(1/(1 + e^{-\mathbf{w}^{(k)\top} \mathbf{x}_n^{(k)}}))$

Automated Curriculum Learning, 2017

Algorithm 1 Intrinsically Motivated Curriculum Learning

Initially: $w_i = 0$ for $i \in [N]$

for $t = 1 \dots T$ **do**

$$\pi(k) := (1 - \epsilon) \frac{e^{w_k}}{\sum_i e^{w_i}} + \frac{\epsilon}{N}$$

Draw task index k from π

Draw training sample $\mathbf{x}$ from D_k

Train network p_θ on $\mathbf{x}$

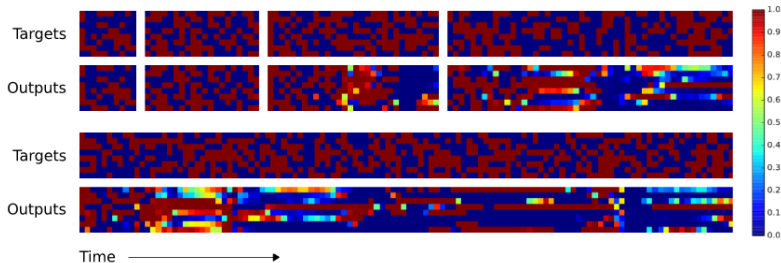
Compute learning progress ν (Sections 3.1 & 3.2)

Map $\hat{r} = \nu / \tau(\mathbf{x})$ to $r \in [-1, 1]$ (Section 2.3)

Update w_i with reward r using Exp3.S (1)

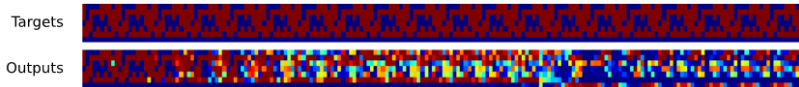
end for

Repeat-copy task, 2014

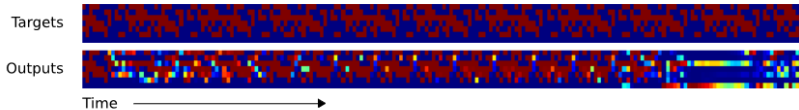


LSTM

Length 10, Repeat 20



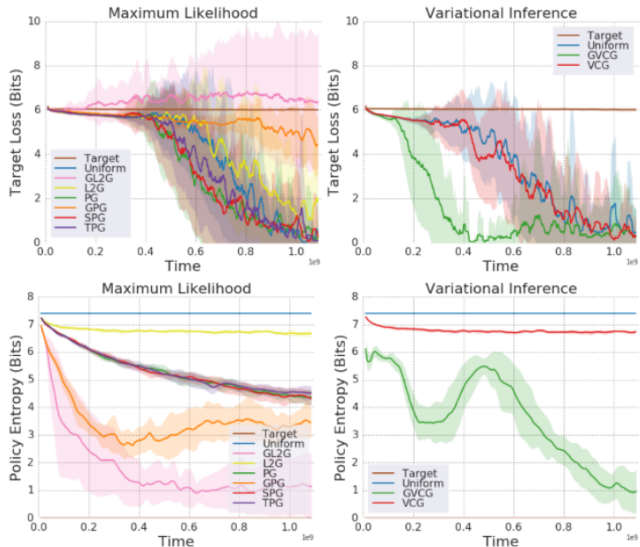
Length 20, Repeat 10



Automated Curriculum Learning, 2017

- Loss-driven Progress
 - ▶ Prediction Gain: $L(\mathbf{w}', \mathbf{x}) - L(\mathbf{w}, \mathbf{x})$
 - ▶ Gradient prediction gain
 - ▶ Self-prediction gain: sampling $\mathbf{x}$
 - ▶ Mean prediction gain: averaging across the tasks
- Complexity-driven Progress
 - ▶ Variational complexity gain: $KL(q|p) - KL(q|p)$
 - ▶ Gradient Variational complexity gain
 - ▶ L2G: difference in l_2 regularization
 - ▶ GL2G: L2G gradient

Automated Curriculum Learning, 2017



Continual learning

Continual learning

Continual Learning is a concept to learn a model for a large number of tasks sequentially without forgetting knowledge obtained from the preceding tasks, where the data in the old tasks are not available any more during training new ones.

Three scenarios for continual learning

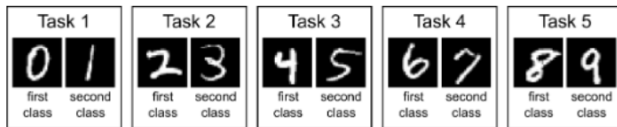
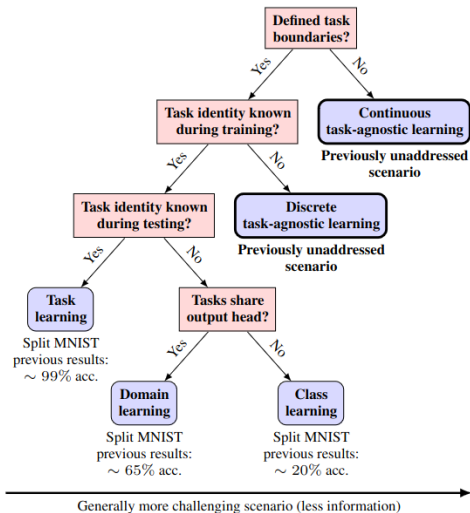


Figure 1: Schematic of split MNIST task protocol.

Table 2: Split MNIST according to each scenario.

Task-IL	With task given, is it the 1 st or 2 nd class? (e.g., 0 or 1)
Domain-IL	With task unknown, is it a 1 st or 2 nd class? (e.g., in [0, 2, 4, 6, 8] or in [1, 3, 5, 7, 9])
Class-IL	With task unknown, which digit is it? (i.e., choice from 0 to 9)

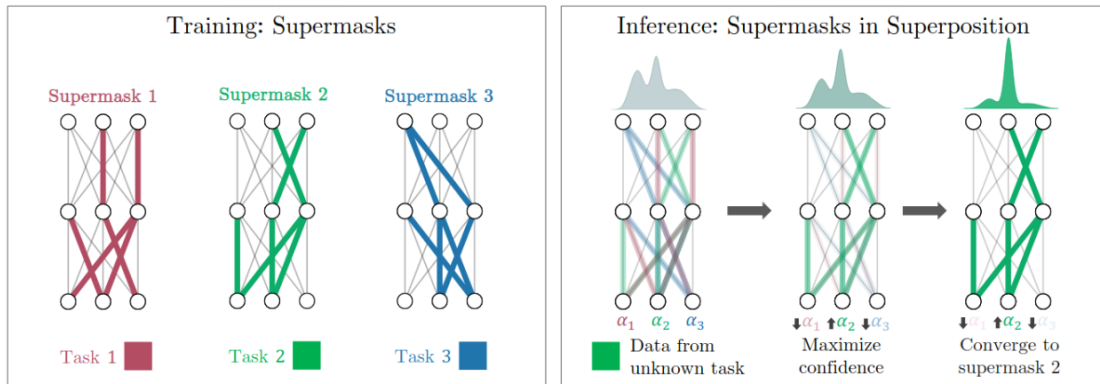
Task Agnostic Continual Learning Using Online Variational Bayes



Continual Learning: task categorization

Scenario	Description	Task space discrete or continuous?	Example methods / task names used
GG	Task G iven during train and G iven during inference	Either	PNN [42], BatchE [51], PSP [4], “Task learning” [55], “Task-IL” [49]
GNs	Task G iven during train, N ot inference; shared labels	Either	EWC [23], SI [54], “Domain learning” [55], “Domain-IL” [49]
GNu	Task G iven during train, N ot inference; u nshared labels	Discrete only	“Class learning” [55], “Class-IL” [49]
NNs	Task N ot given during train N or inference; shared labels	Either	BGD, “Continuous/discrete task agnostic learning” [55]

Supermasks in Superposition¹



¹See talk of Maria Kovaleva, 2023

Multitask learning and inductive bias²

Wiki

Multitask Learning is an approach to inductive transfer that improves generalization by using the domain information contained in the training signals of related tasks as an inductive bias. It does this by learning tasks in parallel while using a shared representation; what is learned for each task can help other tasks be learned better.

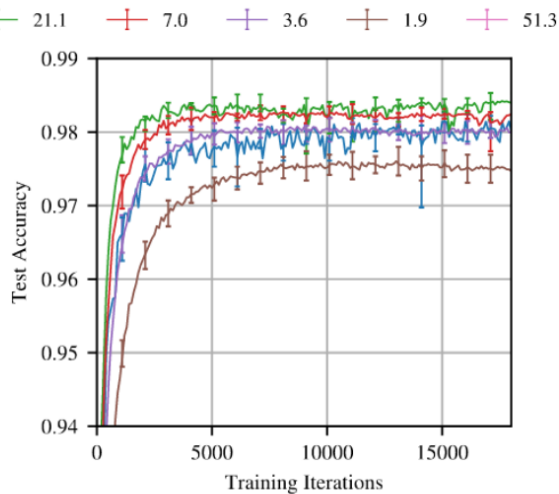
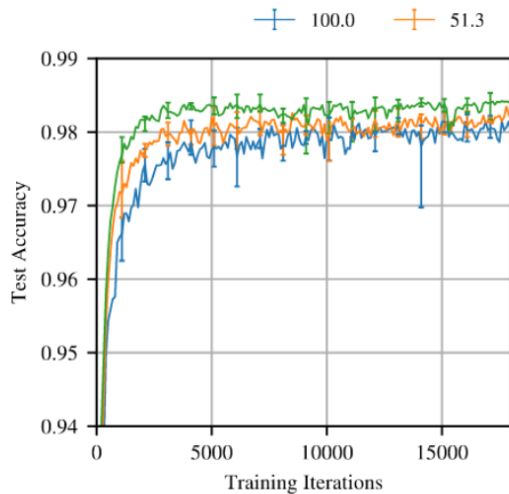
²First paper about this topic: Multitask Learning: A Knowledge-Based Source of Inductive Bias

Lottery ticket

The Lottery Ticket Hypothesis, Frankle et al., 2019

A randomly-initialized, dense neural network contains a subnetwork that is initialized such that—when trained in isolation—it can match the test accuracy of the original network after training for at most the same number of iterations.

Lottery ticket: MNIST



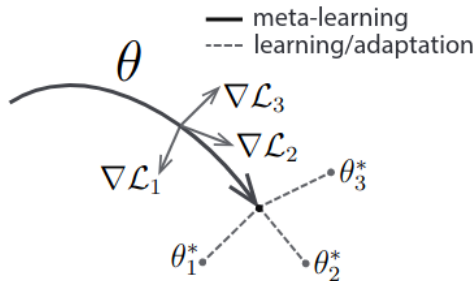
Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks³

Algorithm 1 Model-Agnostic Meta-Learning

Require: $p(\mathcal{T})$: distribution over tasks

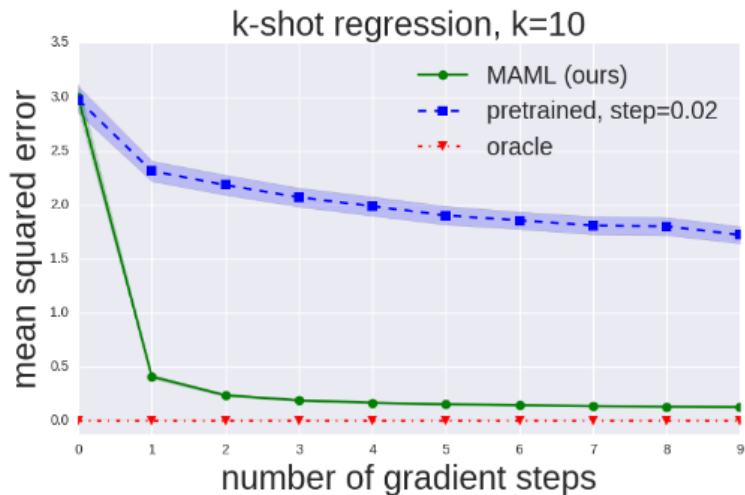
Require: α, β : step size hyperparameters

- 1: randomly initialize θ
 - 2: **while** not done **do**
 - 3: Sample batch of tasks $\mathcal{T}_i \sim p(\mathcal{T})$
 - 4: **for all** $\mathcal{T}_i$ **do**
 - 5: Evaluate $\nabla_{\theta} \mathcal{L}_{\mathcal{T}_i}(f_{\theta})$ with respect to K examples
 - 6: Compute adapted parameters with gradient descent: $\theta'_i = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\mathcal{T}_i}(f_{\theta})$
 - 7: **end for**
 - 8: Update $\theta \leftarrow \theta - \beta \nabla_{\theta} \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(f_{\theta'_i})$
 - 9: **end while**
-

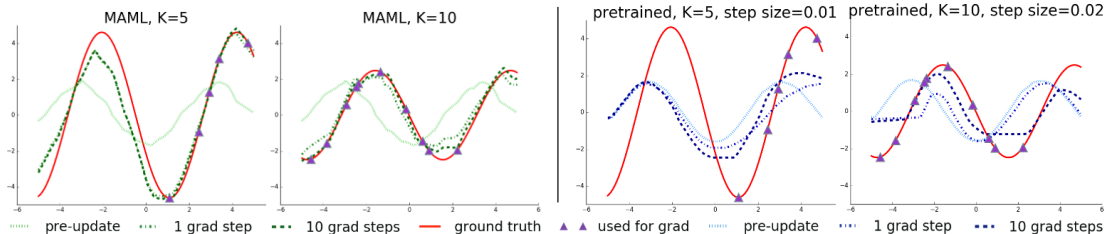


³See also the talk of Dmitriy Protasov on MAML extensions

Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks



Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks



Learning Inductive Biases with Simple Neural Networks

1st-order Generalization Test

"wif" "wif" "wif" "wif"



This is a "wif."



Where is the other "wif?"

1



2



3



2nd-order Generalization Test (the shape bias test)

This is a "dax."



Where is the other "dax?"

1



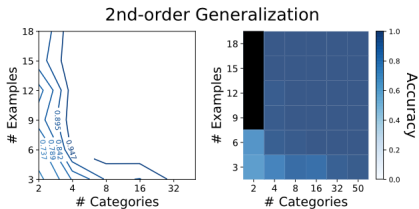
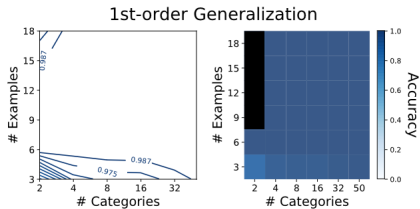
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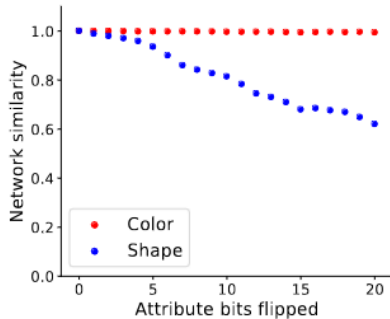
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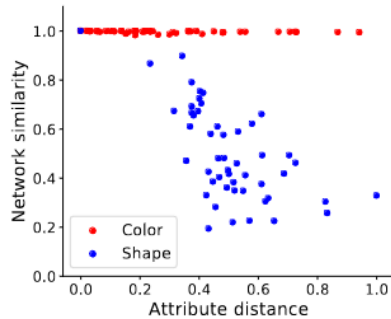
Learning Inductive Biases with Simple Neural Networks



Learning Inductive Biases with Simple Neural Networks

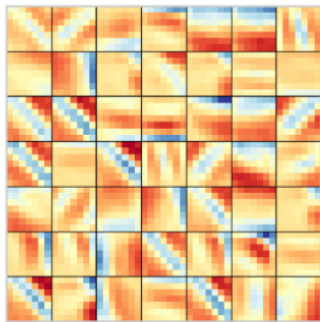


(a) MLP

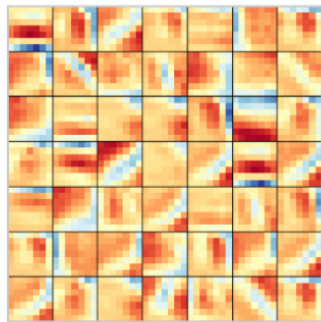


(b) CNN

The deep weight prior: Atanov et al., 2019



(b) Learned filters



(c) Samples from DWP

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